Abhijeet Kumar

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Software Engineer with 4+ years of backend experience, specializing in Golang to build and scale distributed microservices. Delivered production systems powering 15,000+ EV chargers using Kafka, Kubernetes, and event-driven architecture.

Experience

Ford Motor Company Jul 2022 – Present
Software Engineer *Bengaluru, Karnataka*

- Designed and implemented key backend components for the single-tenant to multi-tenant architecture migration,, significantly enhancing system scalability and efficiency, and reducing costs by approximately **40 %**.
- Delivered scalable backend services for the monolith to microservices migration, contributing to service design, implementation, and production deployment.
- Played a key role in migrating **8,200+** EV chargers by designing and implementing scalable OCPP gateway-based microservices, decommissioning **2,150** legacy servers and cutting infrastructure costs by **$9,000** per month.
- Own the end-to-end development and production support of the CDR Cyclic Worker, a high-availability backend service built in Go — leveraging goroutines and worker pools for concurrent Charge Detail Record processing and publishing at scale.
- Designed and implemented the OCPP Gateway microservice features in Go, utilizing goroutines and channels to concurrently handle real-time WebSocket connections and message streams from 15,000+ EV chargers.
- Built and consumed RESTful APIs across Charger Service and Charge Management System, enabling seamless communication between EV charging infrastructure components and upstream platform services.
- Partnered with hardware and firmware teams to deliver and validate cloud features on physical EV chargers, resolving complex integration issues and ensuring OCPP-compliant behavior for new charger models.
- Developed microservices following **OCPP 1.6** and OCPI protocols, and implemented key features from **OCPP 2.0.1** to support protocol upgrades, improving compatibility with newer chargers and enhancing overall system performance and reliability.
- Led the development of major components of the GOS Global OCPP Simulator, implementing critical protocol features and backend services to replicate real EV charging behavior using **OCPP 1.6** and **OCPP 2.0.1**, enabling accurate testing, reducing testing time by 50% and eliminating dependency on physical hardware.
- Owned the Services page of the Operations Console, translating operational pain points from L3 and Support teams into core features that streamlined charger configuration, control, and backend communication.

Intugine Technologies Mar 2022 – Jun 2022
Fullstack Developer Intern *Bengaluru, Karnataka*

- Contributed to the initiative of setting up the open-source Radiant-Design UI project, working with React.js components and tools like Rollup and Webpack, for integration into the frontend projects.
- Built tailored backend features addressing operational and customer support requirements, resolving production issues across multiple product lines.

Intugine Technologies Jun 2021 – Dec 2021
Backend Developer Intern *Bengaluru, Karnataka*

- Developed an automated reporting solution using **JavaScript, Serverless, AWS Lambda, PostgreSQL, AWS S3, and Chart.js**, managing reports for multiple products across over **50** clients. Enabled customizable report generation, significantly saving over **90%** of the time for customer support and operations teams by offering configurable reports tailored to each customer, highlighting logistics statistics and monthly improvements.
- Worked on multiple backend projects, implementing **MQTT** protocols and establishing **AWS SNS** alerts for vehicle threshold notifications.
- Recognized with the **Star Intern Award** for exceptional performance and significant contributions to team projects.

Technical Skills

Languages: Golang, Typescript, Javascript, Python
Frameworks: Nest.js, Express.js, Node.js, React Native, Expo, Flask
Databases: MongoDB, PostgreSQL
Cloud & Messaging: AWS, Docker, Kubernetes, Kafka, Redis
DevOps & Tools: CI/CD, Git, Jira, Datadog, Dynatrace
Concepts: Distributed Systems, Event-Driven Architecture, High Availability Systems, System Design, Data Structures and Algorithms

Projects

BookCave | *Node.js, Express.js, MongoDB, AWS S3, REST API*

- Designed and developed a REST API backend for an online personal library application, enabling users to add, organize, and access their books anywhere.
- Integrated features for uploading books from local storage, adding books via downloadable PDF/EPUB links enhancing user flexibility, and providing an in-app dictionary with Google Dictionary API for real-time word definitions.
- Ensured robust data validation and file type checks using AJV, improving data integrity and security.
- Conducted comprehensive API testing with Supertest, Mocha, and Chai, ensuring reliability and performance.

Education

Dayananda Sagar College Of Engineering 2018 - 2022
Bachelor of Engineering in Computer Science Engineering (GPA: 8.64 / 10.00) *Bengaluru Karnataka*